

Power Stone

Technical Sheet — Participant Guide

Data Space Club — ENSSEA
Global Superstore Dataset (2011 – 2014)

Quick Summary

Prediction: 50% | **Report:** 35% | **Presentation:** 15% | **Bonus:** +5%

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1 The Problem

You are a Data Scientist working for a global retail company. Your mission: **understand the profit** and help the company grow.

Your job has **three parts**:

1.1 Part A — Predict the Profit (50%)

For every transaction (row) in the test set, **predict the Profit value**. This is a **regression** problem — the closer your prediction is to the real profit, the better your score.

1.2 Part B — Business Report (35%)

Using the training data, find:

- Which **Category** is the most profitable and which is the least profitable.
- Inside each Category, use **Sub-Categories** to **explain why**.
- Write a PDF report with your findings + **5 recommendations**.

1.3 Part C — Presentation (15%)

On the final day, present your work in **5 minutes**.

2 Files Provided

File	Description
`train.csv`	Training data (with the `Profit` column)
`test.csv`	Test data (**without** `Profit`) — predict it
`sample_submission.csv`	Submission format template
`technical_sheet.md`	This guide

3 The Dataset

Info

Source: Global Superstore Sales (2011–2014)
Size: 50,000 transactions | **Columns:** 24

Column	Type	Description
`Row ID`	int	Unique row identifier
`Order ID`	str	Unique order ID
`Order Date`	date	When the order was placed
`Ship Date`	date	When the order was shipped
`Ship Mode`	str	Standard / Second / First / Same Day
`Order Priority`	str	Low / Medium / High / Critical

3.1 Order Information

3.2 Customer Information

Column	Type	Description
`Customer ID`	str	Unique customer ID
`Customer Name`	str	Customer name
`Segment`	str	Consumer / Corporate / Home Office
`City`	str	Customer's city
`State`	str	Customer's state
`Country`	str	Customer's country
`Postal Code`	str	Postal code (may have missing values)
`Market`	str	Broad market (e.g. APAC, EU, US)
`Region`	str	Geographic region

3.3 Product Information

Column	Type	Description
`Product ID`	str	Unique product ID
`Category`	str	**Furniture / Office Supplies / Technology**
`Sub-Category`	str	17 values (Chairs, Tables, Phones, etc.)
`Product Name`	str	Full product name

3.4 Sales Information

Important Note

`Profit` is **removed from `test.csv`**. You must predict it.

4 Scoring

Column	Type	Description
`Sales`	float	Total sale amount (USD)
`Quantity`	int	Number of units sold
`Discount`	float	Discount applied (0.0 – 0.85)
`Profit`	float	**TARGET** — Can be negative (loss)
`Shipping Cost`	float	Shipping cost (USD)

4.1 Part A — Prediction Score (50%)

Info

Metric: RMSE (Root Mean Squared Error)
Formula: `RMSE = sqrt(mean((y_true - y_predicted)^2))`

Submission Format — Submit `submission.csv`:

```
Row ID,Predicted Profit 1,123.45 2,-45.67 3,890.12 ...
```

- One row per test row.
- `Row ID` must match the test set.
- `Predicted Profit` is your predicted value (float).

4.2 Part B — Business Report (35%)

Submit a **PDF file** named `report.pdf`:

4.2.1 Section 1: Executive Summary (½ page)

Brief overview + top 3 findings.

4.2.2 Section 2: EDA Insights (1–2 pages)

Distribution of `Profit`, Profit by `Category`, Profit by `Region`/`Market` (with charts).

4.2.3 Section 3: Category Analysis (1–2 pages)

For **each of the 3 Categories**, answer:

1. Is it profitable or losing money? (Total + average profit)
2. Which **Sub-Categories** are the most profitable?
3. Which **Sub-Categories** are the least profitable?
4. **Explain why** — give at least **3 reasons** based on data.

Important Note

Use Sub-Categories to **explain** why the Category is good or bad. Don't just list numbers — tell the story.

4.2.4 Section 4: Recommendations (½–1 page)

Give **5 actionable recommendations**. Each must be: **Specific**, **Data-based**, **Realistic**.

Requirement

Good: "Reduce discounts on Tables from 30% to 15%. Tables generated $X in losses due to high discounts (avg 30%)."

Bad: "Improve marketing." (too vague)

4.2.5 Section 5: Limitations (¼ page)

What couldn't you solve? What assumptions did you make?

Bonus Challenge

Bonus (+5 points): For each Sub-Category, identify the **Top-3** and **Bottom-3** products by total profit. Each product must have at least **20 orders**.

4.3 Part C — Presentation (15%)

On the final day, present in **5 minutes** (3 min presentation + 2 min Q&A).

Info

Cover: ML approach + top business finding + best recommendation.
Judges evaluate: Clarity, storytelling, quality of visuals (1–2 charts max), Q&A.

5 Suggested Workflow

Phase	Task	Time
1	Load data, EDA, understand columns	1h
2	Handle missing values + encode categoricals	1h
3	Build ML model (Linear, RF, XGBoost, LightGBM)	2h
4	Validate with K-Fold CV, tune hyperparameters	1h
5	Predict on test set → `submission.csv`	30m
6	Analyze Categories + Sub-Categories	1.5h
7	Write the PDF report	1.5h
8	Prepare the presentation	1h
Total		**8.5h**

6 Rules

1. **No external data** — use only provided files.

2. **AI tools allowed** for guidance and code help, but you must understand your solution.
3. **No copying** from other teams.
4. **Submission deadline:** Saturday 10:00 AM.
5. The `Profit` column in `test.csv` is **secret** — do not look for it online.
6. You must submit **3 files:** `submission.csv`, `report.pdf`, `notebook.ipynb` or `script.py`.

7 Evaluation Summary

Part	Weight	What's evaluated
Prediction	50%	RMSE on test set
Report	35%	EDA + Category/Sub-Category analysis + Recommendations
Presentation	15%	Clarity + Storytelling + Q&A
Bonus	+5%	Product-level analysis inside Sub-Categories
Total	**100%**	**(+5% bonus)**